Experimental data for N_2 (Nitrogen diatomic)

22 02 02 11 45

Other names
Diatomic nitrogen; Dinitrogen; Molecular nitrogen; Nitrogen; Nitrogen gas; Nitrogen-14;

INChI	INChIKey	SMILES	IUPAC name
InChI=1S/N2/c1-2	IJGRMHOSHXDMSA-UHFFFAOYSA-N	N#N	Dinitrogen

State	Conformation
$^1\Sigma_g$	D*H

Enthalpy of formation (Hfg), Entropy, Integrated heat capacity (0 K to 298.15 K) (HH), Heat Capacity (Cp)

Property	Value	Uncertainty	units	Reference	Comment
Hfg(298.15K) $\Delta_f H°$	0.00	0.00	kJ mol^{-1}	CODATA	
Hfg(0K) $\Delta_f H°$	0.00	0.00	kJ mol^{-1}	CODATA	
Entropy (298.15K) $S°$	191.61		J K^{-1} mol^{-1}	CODATA	
Integrated Heat Capacity (0 to 298.15K) $H°-H°(T)$	8.67		kJ mol^{-1}	CODATA	
Heat Capacity (298.15K) $C_p°$	29.12		J K^{-1} mol^{-1}	Gurvich	

Information can also be found for this species in the NIST Chemistry Webbook

Vibrational levels (cm^{-1})

Mode Number	Symmetry	Frequency			Intensity			Comment	Description
		Fundamental(cm^{-1})	Harmonic(cm^{-1})	Reference	(km mol^{-1})	unc.	Reference		
1	Σ_g	2330	2359	2007Iri:389					

Detailed diatomic data

ω_e	$\omega_e x_e$	$\omega_e y_e$	B_e	α_e	ZPE	reference
2358.57	14.324	-0.00226	1.998241	0.017318	1175.778	2007Iri:389

vibrational zero-point energy: 1165.0 cm^{-1} (from fundamental vibrations)
Calculated vibrational frequencies for N_2 (Nitrogen diatomic).

Rotational Constants (cm^{-1})

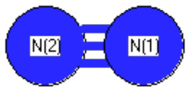
See section I.F.4 to change rotational constant units

A	B	C	reference	comment
	1.99824		1979HUB/HER	

Calculated rotational constants for N_2 (Nitrogen diatomic).

Product of moments of inertia $I_A I_B I_C$				
8.436234	amu Å^2		1.400892E-39	gm cm^2

Geometric Data



Point Group $D_{\infty h}$

Internal coordinates

distances (r) in Å, angles (a) in degrees, dihedrals (d) in degrees

Description	Value	unc.	Connectivity				Reference	Comment
			Atom 1	Atom 2	Atom 3	Atom 4		
rNN	1.098		1	2			1979HUB/HER	re

Cartesians

Atom	x (Å)	y (Å)	z (Å)
N1	0.0000	0.0000	0.5488
N2	0.0000	0.0000	-0.5488

Atom - Atom Distances

Distances in Å

	N1	N2
N1		**1.0977**
N2	**1.0977**	

Calculated geometries for N_2 (Nitrogen diatomic).

Bond descriptions

Examples: C-C single bond, C=C, double bond, C#C triple bond, C:C aromatic bond

Bond Type	Count
N#N	1

Connectivity

Atom 1	Atom 2
N1	N2

Electronic energy levels (cm^{-1})

Energy (cm^{-1})	Degeneracy	reference	description
0	1	1979HUB/HER	$^1\Sigma_g$
59619.3	6	1979HUB/HER	$^3\Pi$

Ionization Energies (eV)

Ionization Energy	I.E. unc.	vertical I.E.	v.I.E. unc.	reference
15.581	0.008	15.581		webbook

Proton Affinity (kJ mol^{-1})

Proton Affinity	unc.	Product	reference	comment
493.8		NNH^+	webbook	

Dipole, Quadrupole and Polarizability

Electric dipole moment

State	Config	State description	Conf description	Exp. min.	Dipole (Debye)				Reference	comment	Point Group	Components	
					x	y	z	total				dipole	quadrupole
1	1	$^1\Sigma_g$	$D_{\infty h}$	True				0.000	NSRDS-NBS10		$D_{\infty h}$	0	1
2	1	$^3\Pi$	$D_{\infty h}$	False				0.000	_		$D_{\infty h}$	0	1

Experimental dipole measurement abbreviations: MW microwave; DT Dielectric with Temperature variation; DR Indirect (usually an upper limit); MB Molecular beam

Calculated electric dipole moments for N_2 (Nitrogen diatomic).

Electric quadrupole moment

State	Config	State description	Conf description	Exp. min.	Quadrupole (D Å)			Reference	comment	Point Group	Components	
					xx	yy	zz				dipole	quadrupole
1	1	$^1\Sigma_g$	$D_{\infty h}$	True	0.697	0.697	-1.394	1998Gra/Imr:49	-4.65+-0.08 E-40 C m^2	$D_{\infty h}$	0	1
2	1	$^3\Pi$	$D_{\infty h}$	False						$D_{\infty h}$	0	1

Calculated electric quadrupole moments for N_2 (Nitrogen diatomic).

Electric dipole polarizability ($Å^3$)

alpha	unc.	Reference
1.710		1997Oln/Can:59

Calculated electric dipole polarizability for N_2 (Nitrogen diatomic).

References

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squib	reference	DOI
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1997Oln/Can:59	TN Olney, NM Cann, G Cooper, CE Brion, Absolute scale determination for photoabsorption spectra and the calculation of molecular properties using dipole sum-rules, Chem. Phys. 223 (1997) 59-98	10.1016/S0301-0104(97)00145-6
1998Gra/Imr:49	C Graham, DA Imrie, RE Raab "Measurement of the electric quadrupole moments of CO2, CO, N2, Cl2 and BF3" Mol. Phys. 93(1), 1998, 49-56	10.1080/002689798169429
2007Iri:389	KK Irikura "Experimental Vibrational Zero-Point Energies: Diatomic Molecules" J. Phys. Chem. Ref. Data 36(2), 389, 2007	10.1063/1.2436891
CODATA	Cox, J.D.; Wagman, D.D.; Medvedev, V.A.CODATA Key Values for Thermodynamics. Hemisphere, New York, 1989	10.1002/bbpc.19900940121
Gurvich	Gurvich, L.V.; Veyts, I. V.; Alcock, C. B., Thermodynamic Properties of Individual Substances, Fouth Edition, Hemisphere Pub. Co., New York, 1989	
NSRDS-NBS10	R. D. Nelson Jr., D. R. Lide, A. A. Maryott "Selected Values of electric dipole moments for molecules in the gas phase" NSRDS-NBS10, 1967	10.6028/NBS.NSRDS.10
webbook	NIST Chemistry Webbook (http://webbook.nist.gov/chemistry)	10.18434/T4D303

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TiN (Titanium mononitride)	N2- (nitrogen diatomic anion)